

CLOUDFLARE RADAR

Cloudflare Radar 2024 Year in Review

Internet patterns and trends observed through Cloudflare's global network during 2024

Audience: Technology Executives & Network Infrastructure Leaders

Data Source: Cloudflare Network | radar.cloudflare.com/year-in-review/2024

17% Global Traffic Growth and Starlink's 3.3x Surge

+17%

YoY Global Traffic Growth

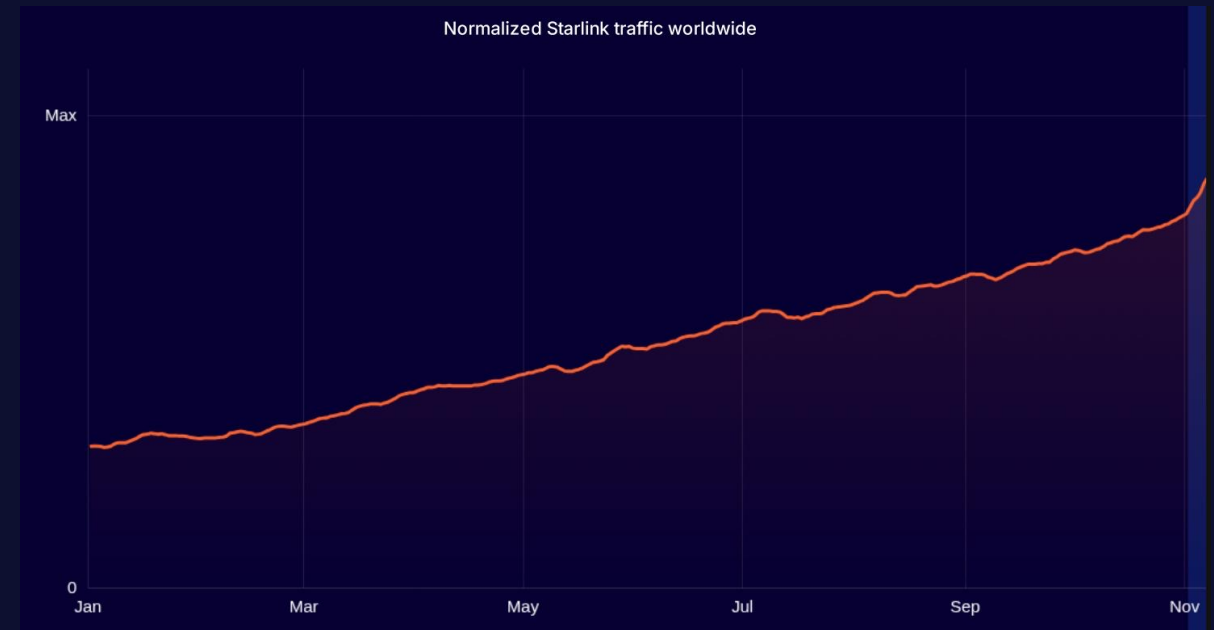
3.3x

Starlink Traffic Surge

41%

Mobile Device Traffic

- Global Internet traffic grew 17% YoY, driven by communication, commerce, entertainment & transportation
- Trend line starts mid-January to normalize for post-holiday activity
- SpaceX Starlink expanded satellite connectivity to underserved regions



Google Leads Top 10 Services as Generative AI Rises

- #1 Google
- #2 Facebook
- #3 Apple
- #4 TikTok
- #5 AWS
- #6 Microsoft
- #7 Instagram
- #8 YouTube
- #9 Amazon
- #10 WhatsApp

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

Ranking based on aggregate 1.1.1.1 DNS resolver data • Generative AI continued rapid rise • Metaverse receded

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

2% →

Post-Quantum TLS 1.3
13%

21%

HTTP/3 Adoption

Go #1

API Client Language

Post-Quantum Encryption

- Chrome 124 enabled PQ key agreement by default (Apr 17)
- Adoption surged from ~2% to ~13% of TLS 1.3 traffic
- Firefox rolling out PQ; Safari began initial testing

Additional Adoption Metrics

- Mobile OS: 33% iOS; Android peaks >95% in some regions
- Search: Google 89%, Yandex 3.1%, Baidu 2.7%, Bing 2.6%
- API Languages: Go ~12%, then Node.js, Python 9.6%, Java 7.4%, .NET 3.6%
- Browser: Chrome remains #1 globally

Post-quantum encrypted traffic worldwide



225 Major Disruptions: Government Shutdowns Lead Outage

Causes

225

Major Disruptions

116+

Govt-Directed Shutdowns

41

Cable Cuts



Government Directed (116+)

Cable Cut (41)

Unknown (16)

Power Outage (16)

Technical Problem (9)

Military Action (9)

Weather (8)

Maintenance (4)

Cyber Attack (4)

Natural Disaster (1)

Fire (1)

Government Directed

Cable Cut

Unknown

Power Outage

Technical Problem

Military Action

Weather

Maintenance

41

16

16

9

9

8

4

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

6.5%

Traffic Mitigated

68%

Bot Traffic from Top 10
Countries

4.3%

Malicious Emails

Attack Surface

- Gambling & Games: Most attacked industry sector
- Log4j (CVE-2021-44228): Still actively exploited 2+ years post-disclosure
- RPKI valid routes increased 6.4% — improving routing security

Email Threat Landscape

- 4.3% of emails classified as malicious
- 43% of malicious emails contained deceptive links
- .bar TLD: Largest share of malicious & spam email origin

Bot Concentration

- 68% of global bot traffic from just the top 10 countries

Log4j persists

Gambling/Games #1 target

.bar TLD top spam origin

+6.4% RPKI valid routes

Five Takeaways Shaping the 2025 Internet Outlook

1

Traffic Growth Trajectory

Global traffic +17% YoY; Starlink 3.3x surge extends connectivity to underserved regions — plan for sustained bandwidth demand.

2

Post-Quantum Momentum

PQ encryption leapt 2% → 13% of TLS 1.3 after Chrome 124 default — accelerate PQ readiness in enterprise TLS stacks.

3

Connectivity Fragility

Government-directed shutdowns led 225 disruptions — build resilience & multi-path architectures for politically volatile regions.

4

Persistent Security Threats

6.5% traffic mitigated; Log4j still exploited 2+ years on — prioritize patching velocity & assume perpetual threat exposure.

5

Evolving Developer Ecosystem

Go became #1 API client language; HTTP/3 at 21% — align infrastructure investment with modern protocol & toolchain adoption.